

Curriculum Vitae of JIALE CHENG

3rd year Ph.D. Candidate

Electrical Engineering and Computer Science, the University of Michigan, Ann Arbor MI-48105

Email: jlcheng@umich.edu

Personal Website: sites.google.com/umich.edu/jialecheng

CURRENT RESEASARCH INTEREST

Graph alignment theory, Information theory, and there applications to networks and systems

EDUCATION

Southeast University (SEU), Chien-Shiung Wu Honored College Nanjing, China
Bachelor(Hons.) of Engineering in Communication Engineering 09/2016-06/2020

Supervisor: Prof. Xiaohan Sun

Research Fields: Signal processing, MIMO-OFDM, communications, edge computing, reconfigurable intelligent surface

Southeast University (SEU), School of Information Science and Engineering Nanjing, China
Master of Science in Communications 09/2020-06/2023

Supervisor: Prof. Nan Liu, Prof. Wei Kang

Research Fields: information theory, network coding, private information retrieval, coded caching, secure multiparty computation

University of Michigan, Electrical Engineering and Computer Science Ann Arbor, U.S.
Doctor of Philosophy (pursuing) in Electrical and Computer Engineering (ECE) 08/2023-Present

Supervisor: Prof. Lei Ying

Research Fields: network alignment, graph inferences

PUBLICATIONS

In preparation:

- [1]. **J. Cheng**, Z. Wang and L. Ying, "Galton-Watson tree testing with diverging offsprings with application to Graph Matching", in preparation.
- [2]. **J. Cheng**, H. Tong and L. Ying, "Network Alignment with Positively and Negatively Correlated Edges", in preparation.

Preprints:

- [3]. Q. Lu, **J. Cheng**, W. Kang and N. Liu, "Capacity of Hierarchical Secure Coded Gradient Aggregation with Straggling Communication Links", in ArXiv preprint, 2024, arXiv: 2412.11496. [Get paper](#)

Published:

- [4]. **J. Cheng**, N. Liu, W. Kang and Y. Li, "The Capacity of Symmetric Private Information Retrieval Under Arbitrary Collusion and Eavesdropping Patterns," in *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 3037-3050, 2022, doi: 10.1109/TIFS.2022.3201375. [Get paper](#)
- [5]. **J. Cheng**, N. Liu, W. Kang, "On the Asymptotic Capacity of Information Theoretical Privacy-preserving Epidemiological Data Collection," in *Entropy (Special Issue: Advances in Information and Coding Theory)*, vol. 25, no. 4:625, 2023, doi: 10.3390/e25040625. (Featured Paper) [Get Paper](#).

CONFERENCE PARTICIPATION

2022 IEEE East Asian School of Information Theory (EASIT2022)

Shenzhen, China	08/2022
Participated as a student member	
Student Poster Session: “The Capacity of Symmetric Private Information Retrieval Under Arbitrary Collusion and Eavesdropping Patterns”	

2021 IEEE East Asian School of Information Theory (EASIT2021)

Online	08/2021
Participated as a student member	

2019 IEEE International Workshop on Signal Processing Systems (SiPS2019)

Nanjing, China	10/2019
Student Poster Session & Oral Presentation: “The Design and Simulation of Phased Array Focus in Ultrasonic Wireless Power Transmission”	

2017 American Association of Physics Teachers (AAPT) Summer Meeting

Cincinnati, OH	07/2017
Student Poster Session: “On the falling of a rigid body”	

RESEARCH EXPERIENCE BEFORE UMICH

National Mobile Communications Research Laboratory (NCRL)

Nanjing, China	09/2020-Present
Interests in information theory and network coding	
Participate in the research supported by the National Natural Science Foundation of China	
Published a paper in IEEE Transactions on Information Forensics and Security	
Filed a Chinese patent (Publication No. CN 114567488 A)	

National Engineering Research Center for Optical Sensor/Communication Integrated Network

Nanjing, China	09/2018-02/2020
Using the Phased Array Focus to study the Ultrasonic Wireless Power Transmission	
Responsible for the simulation of a proposed design before physical implementation	
Participated in SiPS2019	

HONORS & AWARDS

The Rackham Doctoral Student Fellowship	2023
Honorable Mention in Mathematical Contest of Modeling (ICM)	2019
National Endeavour Fellowship of China, Southeast University	2017

SKILLS & INTERESTS

Language Skills: Chinese (native), English (fluent), Japanese (elementary)

Computer Skills: MATLAB, C/C++, Microsoft Office Tools, LaTeX, Python, Mathematica, Photoshop

Interests: Reading, Piano, Table Tennis